
Working Memory in the Chain of Thought Is Not Captured by J-Space

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Abstract

1 A common reading of J-space, the concept workspace a fitted Jacobian lens reads
2 from the residual stream, is that reasoning state is carried internally and the chain
3 of thought externalizes state the workspace can no longer hold. Behavioral mea-
4 surements from 1.5B to 671B parameters show no sign of such relief. Instructed to
5 answer chained arithmetic without writing, models fail beyond one to two depen-
6 dent steps. In free generation, models write 0.93 to 1.00 of intermediate values
7 from one-step problems on, so writing has no onset threshold. No intervention
8 moves state across the boundary: damaging the residual stream does not increase
9 writing, capping writing does not move computation inward, and ablating the top
10 active J-space concepts at the strongest dose that preserves ordinary generation
11 changes nothing against a size-matched random control. We then locate the mem-
12 ory that does carry the state, with causal interventions on six unmodified models
13 from four families (4B to 14B). The hidden state under a written number holds the
14 value later steps use. Overwriting it with the state a different number would have
15 produced, text unchanged, redirects the final answer to that never-written number
16 in 0.70 to 1.00 of runs; size-matched random noise redirects at most 0.01 on the
17 four instruction-tuned models. Later steps retrieve the value through attention to
18 its position, the retrieval completing by layer 23 of 36 on Qwen3-4B. Two writ-
19 ten positions hold values independently, and planted pairs compose into a sum
20 appearing nowhere in the input (0.88 and 0.90). Identical text can therefore ac-
21 company different stored values, and a monitor that reads only the text cannot tell
22 them apart.

23 1 Introduction

24 The chain of thought is the primary working memory of serial reasoning in language models: denied
25 writing, models fail within one to two dependent steps, and no intervention we test shifts state
26 between the internal workspace and the written trace. The behavioral claims are established at 1.5B
27 to 671B parameters; the mechanism, with causal interventions on six unmodified models from four
28 families at 4B to 14B, on two task families with exact ground truth for every intermediate value.

29 The capacity half addresses the overflow picture of externalization: reasoning state is carried inter-
30 nally, and writing takes over when the workspace can no longer hold it. This is a natural reading of
31 workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens
32 reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show
33 up in three places; it shows up in none.

34 1. **No fallback** (behavioral, 1.5B to 671B). Instructed to answer chained arithmetic with no
35 writing, models fail beyond one to two dependent steps (Section 3).

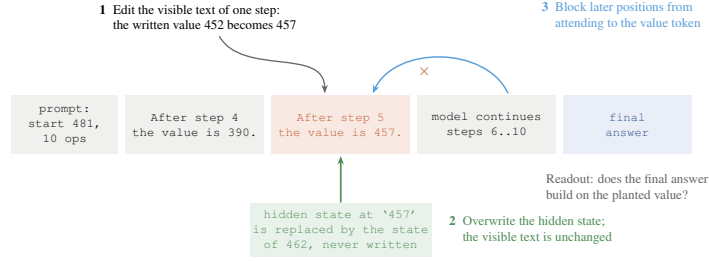


Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

2. **No threshold** (behavioral, 1.5B to 671B). Models write nearly all ground-truth intermediates from one-step problems onward (Section 3).
3. **No reallocation** (behavioral and ablation, 4B to 7B). Damaging the residual stream does not increase writing, capping writing does not move computation inward, and a generation-preserving J-space ablation changes nothing (Section 3).
4. **Storage** (causal, 4B to 14B). The hidden state under a written number carries the value the model later uses: planting a never-written value there, text unchanged, redirects the answer on all six models (Section 4).
5. **Retrieval** (causal, 4B to 14B). Later steps read the value through attention to the written number, in the middle third of the network’s layers (Section 5).
6. **Composition and trust** (causal and behavioral). Two positions are independently addressable, and planted values compose into their sum. Reliance on the trace tracks whether the model can regenerate the state, a candidate explanation for conflicting reports on how trace reliance scales (Sections 4 and 6).

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether an internal alternative exists, how the value is retrieved, or whether stores compose.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events and surface forms. In both, the model writes one line per step giving the running value. A third task, box tracking, updates the contents of several containers in parallel; it appears only as the internally held comparison task for the lesion test in Section 3. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

2.2 Edit-and-continue protocol

The protocol takes a correct worked solution, changes the value at one step j , truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap

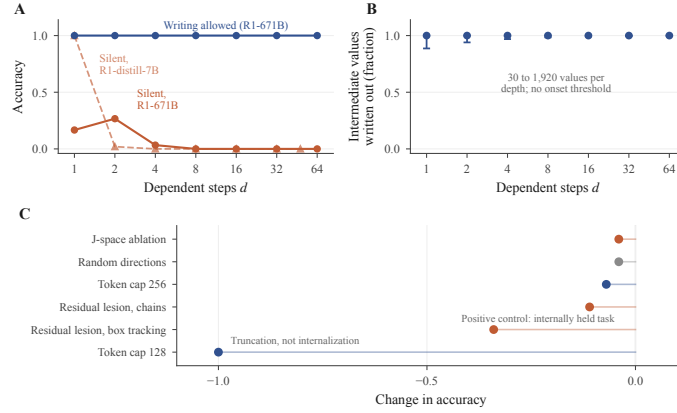


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); no onset threshold appears. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

intervals over items. Appendix C summarizes generation and intervention configuration; prompts, model revisions, and per-run data will be released with the code.

3 Does writing replace internal working memory?

3.1 Performance without written intermediates

Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at $d = 4$; DeepSeek-R1-671B: at most 0.27 at any depth, 0.03 at $d = 4$; DeepSeek V3.2 on modular-arithmetic chains: 0.23 at $d = 1$, at most 0.03 beyond). The values are random, so nothing can be recalled; the failures bound performance under this instruction, with its prompting and training-distribution effects, rather than latent internal capacity.

3.2 When models write intermediate values

Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through $d = 64$, at every scale tested: no difficulty threshold at which writing begins is observed. This rules out an inference-time switch from internal holding to writing; an always-write policy learned in training could itself descend from capacity limits.

3.3 Tests for reallocation between internal and written state

A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held internally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step. Ablating the top 16 active J-space concepts at every position, at the strongest dose preserving ordinary generation (next-token agreement 0.87, perplexity ratio 1.12), leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged on synthetic chains and on GSM8K, doing no more than a size-matched random-directions ablation. The chain-of-thought-versus-direct ablation asymmetry reported for J-space [Gurnee et al., 2026] therefore does not replicate at a coherence-safe dose on our tasks, and the null holds at one to two steps, where models do perform the arithmetic internally. The null bounds this readout at generation-preserving doses rather than the workspace as

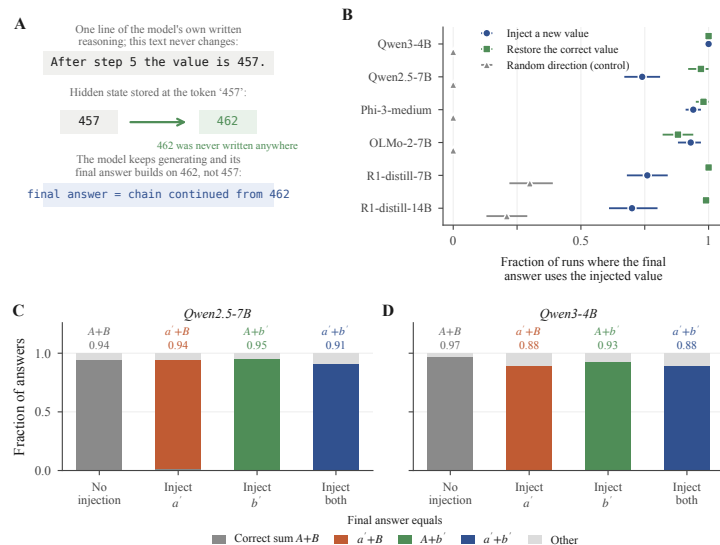


Figure 3: The hidden state under a written number carries the value the model later uses, and two planted values compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, raising their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

98 a whole (Section 8); Section 4 gives the gap an existence proof: a value planted in the hidden state
 99 under a written number is causally decisive, sits in residual states the lens samples, and survives
 100 removal of the top 16 concepts.

101 4 Written tokens store causally active intermediate values

102 4.1 Restoring and replacing stored values

103 Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while
 104 the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting
 105 a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The
 106 size-matched random control sits at or below 0.01 for the four instruction-tuned models. The patch
 107 replaces the position's full contextual state rather than an isolated numeric component; because
 108 answers track whichever value's state is planted, the value that state carries is the causal quantity,
 109 and because the planted value is a mid-chain intermediate the model never produced, the patch does
 110 not inject the answer. The effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B). Rates are
 111 conditioned on items where the text edit was followed (0.44 to 0.98 by model; Appendix A).

112 On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75,
 113 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the
 114 answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

115 4.2 Independent storage and composition

116 In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one
 117 final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.88 and 0.93
 118 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the
 119 single-position rates (0.89 and 0.82). The random patch leaves the clean answer in place (0.96 and

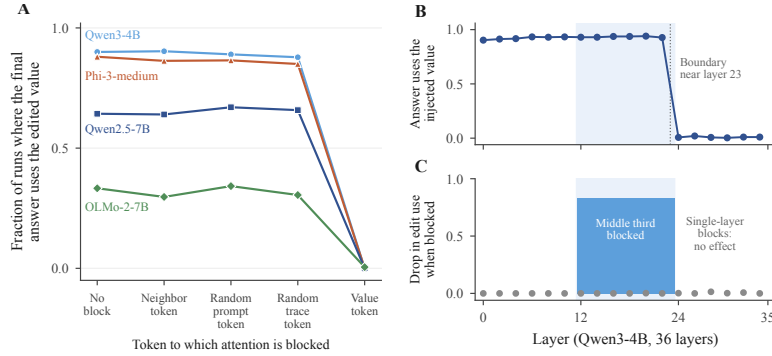


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing. (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking attention across the shaded middle band removes most edit use; no single layer’s block matters (gray dots).

0.95), and answers using only one of the two planted values occur at 0.01. The reported number must be assembled at answer time from two values set independently and never written anywhere.

5 How models retrieve written state

5.1 Attention-blocking experiments

Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s dependence on that value on every model tested (Figure 4A); matched blocks of the adjacent words, a random prompt position, or a random trace position leave it at the unblocked level. Blocking the operand and the next step needs breaks the arithmetic itself, a positive control showing the mask disrupts computation it targets. The collapse holds across three families and both task formats (natural-language: 0.730 to 0.007 and 0.720 to 0.003 on Qwen3-4B and Phi-3-medium). Under the block the model produces a fluent, parseable number matching neither value (unparseable rate 0.000; OLMo-2-7B 0.040, its unblocked level): it computes forward from state it can no longer retrieve. Reading is the preferred path rather than the only one: with the value’s characters replaced by unreadable dots and attention untouched, the Qwen models reconstruct the value from the previous line in 0.60 to 0.73 of cases, Phi-3-medium never.

5.2 Where retrieval occurs across layers

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (Appendix B) show the same structure with model-specific depth.

6 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value whose derivation the prompt states is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly. Llama 3.3 70B follows edits at 0.97 either way, despite solving the same chains from

151 scratch: the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never
 152 revert. The conditions are separate experiments; we claim within-experiment contrasts only.

153 On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10
 154 by V3.2: a benchmark intermediate is often one operation from numbers still in the prompt. Within
 155 one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham
 156 et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks;
 157 our contrasts offer a candidate explanation, that the direction depends on the state’s regenerability.
 158 On regenerable state, stronger models depend less on the trace; on state no model can regenerate,
 159 dependence is high at every scale. Trace reliance here is a separate quantity from verbalization
 160 faithfulness, whether the trace reports the computation that produced the answer; our claims concern
 161 reliance on already-written state.

162 7 Related work

163 Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal
 164 abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate
 165 a component mattering with deleting information at a position, which the single-layer sweep in
 166 Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token
 167 results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally. Wang
 168 [2026] corrupt surface traces only in aggregate and conclude reasoning is latent; our token-level
 169 edits show written state is causally decisive when the model cannot regenerate it, which bounds that
 170 conclusion. Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight
 171 channel; our results identify what such a monitor reads and what it cannot.

172 8 Limitations

173 Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans,
 174 and richer multi-entity state are untested. The J-space null bounds one fitted lens at generation-
 175 preserving doses: stronger doses damage generation nonspecifically, we do not measure the J-space
 176 content of the written positions directly, and the workspace exceeds the lens’s domain by construc-
 177 tion (attention and MLP interiors, the KV cache, and residual directions outside the fitted dictionary).
 178 The fine layer sweeps cover one model, and the coarse sweeps do not resolve Qwen2.5-7B’s read
 179 boundary. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning
 180 on edit-following makes cross-model levels not directly comparable; Appendix A reports the con-
 181 ditioning rates. A structured patch control planting a valid activation for a different quantity would
 182 isolate value content beyond the size-matched noise control used here.

183 9 Conclusion

184 Denied writing, models fail within one to two dependent steps at every scale tested, and no interven-
 185 tion shifts state between workspace and trace: the chain of thought is working memory, settable by
 186 intervention, retrieved through middle-layer attention, independently addressable, and decisive for
 187 the answer. The divergences between text and stored value shown here are experimenter-induced;
 188 the results establish that the representational room for divergence exists and that a monitor reading
 189 only the text cannot detect it [Korbak et al., 2025]. Whether it arises naturally is untested.

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215 A Per-model edit and patch rates

216 Table 1 gives the per-model rates behind Figure 3B, with the text-edit-following rate that conditions
 217 the patch columns. Ten-step chains; brackets are 95 percent bootstrap intervals; n counts items
 218 where the text edit was followed. Unconditioned per-run outcomes are in the released data. The
 219 distill models re-solve the problem after their reasoning phase, which elevates their random-control
 220 reversion: in a text-logged sample, 24 of 24 reverting random-control continuations re-derive from
 221 the prompt’s starting value after the reasoning phase, and none correct the value in place. Re-solving
 222 can only produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is
 223 excluded (12 usable items; random control exceeds the planted-value rate).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

224 B Coarse cross-model layer sweeps

225 The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs
 226 following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers
 227 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the
 228 fine sweep’s structure: the planted value takes effect at any band before the read and at none after.
 229 Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary
 230 is not resolved at this granularity.

231 C Generation and intervention configuration

232 Generation uses temperature 0.6, top- p 0.95, five samples per item. The text edit changes only the
 233 visible token. The state patch overwrites the residual stream at the value’s token position across a
 234 middle band of layers with activations captured from a run in which the different value was written
 235 there. The attention block masks all attention from post-edit positions to the value’s token at all
 236 layers. Sample sizes are 100 to 600 per condition with two to three seeds. Prompts, model revisions,
 237 and per-run configuration files are released with the code and data.